

Papers 2 & 3: Key Topics, Formulae & Worked Examples

A focused academic revision resource for the GCSE Higher calculator papers. Each topic gives the essential formulae followed by fully worked examples demonstrating complete method. Edexcel-aligned; suitable for AQA and OCR.

This guide concentrates on the high-value topics for the calculator papers. For each, the key formulae are stated first, then worked examples model the full method an examiner expects to see. Work through each example yourself before reading the solution. Keep your calculator in **Degrees** mode and carry exact values forward rather than rounding early.

1 Trigonometry — Right-Angled, Sine & Cosine Rules

The calculator papers test trigonometry heavily, both in right-angled triangles and in non-right-angled triangles via the sine and cosine rules. Always identify which rule applies before starting.

KEY FORMULAE

Right-angled: $\sin \theta = \frac{\text{opp}}{\text{hyp}}$ $\cos \theta = \frac{\text{adj}}{\text{hyp}}$ $\tan \theta = \frac{\text{opp}}{\text{adj}}$

Sine rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine rule: $a^2 = b^2 + c^2 - 2bc \cdot \cos A$ (for a side) $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ (for an angle)

Area of a triangle: $\text{Area} = \frac{1}{2} \cdot ab \cdot \sin C$

WORKED EXAMPLE 1 — COSINE RULE FOR A SIDE

A triangle has sides $b = 9.4$ cm and $c = 7.1$ cm with the included angle $A = 68^\circ$. Find the length of side a , to 3 significant figures.

Use the cosine rule because two sides and the included angle are known:

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$a^2 = 9.4^2 + 7.1^2 - 2(9.4)(7.1) \cdot \cos 68^\circ$$

$$a^2 = 88.36 + 50.41 - 133.48 \times 0.37461\dots$$

$$a^2 = 138.77 - 50.003\dots = 88.767\dots$$

$$a = \sqrt{88.767\dots} = 9.4216\dots$$

Keep the full value on your calculator; round only at the end.

$a = 9.42$ cm (3 s.f.)

WORKED EXAMPLE 2 — SINE RULE FOR AN ANGLE, THEN AREA

In triangle PQR, PQ = 11 cm, QR = 8 cm and angle PQR = 52°. Find angle QPR (3 s.f.), then the area of the triangle.

Side QR (8 cm) is opposite the unknown angle QPR; side PQ (11 cm) is opposite angle PRQ. First find the third side or use the sine rule once an opposite pair is known. Here, use the cosine rule first to find PR, since we have two sides and the included angle Q:

$$PR^2 = 11^2 + 8^2 - 2(11)(8) \cdot \cos 52^\circ = 121 + 64 - 176 \times 0.61566... = 185 - 108.36... = 76.64...$$

$$PR = \sqrt{76.64...} = 8.7546... \text{ cm}$$

Now use the sine rule to find angle QPR (opposite QR = 8):

$$\frac{\sin(QPR)}{8} = \frac{\sin 52^\circ}{8.7546...}$$

$$\sin(QPR) = \frac{8 \times \sin 52^\circ}{8.7546...} = \frac{8 \times 0.78801...}{8.7546...} = 0.72009...$$

$$QPR = \sin^{-1}(0.72015...) = 46.06...^\circ$$

Area, using two sides and the included angle Q:

$$\text{Area} = \frac{1}{2} \times 11 \times 8 \times \sin 52^\circ = 44 \times 0.78801... = 34.67... \text{ cm}^2$$

Angle QPR = 46.1° (3 s.f.); Area = 34.7 cm² (3 s.f.)

WORKED EXAMPLE 3 — BEARINGS (GRADE 8/9)

A ship sails 12 km on a bearing of 050°, then changes course and sails 18 km on a bearing of 140°. Calculate its distance from the starting point (3 s.f.) and the bearing on which it must sail to return directly to the start.

The two bearings differ by $140^\circ - 50^\circ = 90^\circ$, so the path turns through a right angle — the two legs are perpendicular. Use Pythagoras for the direct distance d:

$$d = \sqrt{(12^2 + 18^2)} = \sqrt{(144 + 324)} = \sqrt{468} = 21.633... \text{ km}$$

To find the return bearing, first find the angle at the second turning point between the 18 km leg and the direct return path:

$$\tan \theta = \frac{12}{18} \rightarrow \theta = \tan^{-1}(0.6667) = 33.69^\circ$$

The outward second leg was on bearing 140°, so the reverse direction is $140^\circ - 180^\circ = 320^\circ$ (i.e. travelling back along it). The return path lies θ off this; constructing the geometry, the bearing back to start is:

$$\text{Return bearing} = 320^\circ - 33.69^\circ = 286.31^\circ \approx 286^\circ$$

Always draw the diagram and mark North lines at each point — bearings marks are lost on setup, not calculation.

Distance = 21.6 km (3 s.f.); return bearing $\approx 286^\circ$

2 Percentages, Compound Interest & Reverse Percentages

Multi-step percentage problems are a staple of the calculator papers. Use multipliers throughout — they make compound changes and reversals far cleaner than working out each step separately.

KEY FORMULAE

Multiplier for +r%: $1 + \frac{r}{100}$ · **for -r%:** $1 - \frac{r}{100}$

Compound interest / growth: $\text{Final} = P \times (\text{multiplier})^n$

Reverse percentage: $\text{Original} = \frac{\text{Final amount}}{\text{multiplier}}$

Percentage change: $\frac{\text{change}}{\text{original value}} \times 100$

WORKED EXAMPLE 1 — COMPOUND INTEREST WITH COMPARISON

Aisha invests £4500 in an account paying 3.2% compound interest per year. Calculate the value after 4 years (to the nearest penny) and the total interest earned.

Multiplier for +3.2% is 1.032. Apply it 4 times using a power:

$$\text{Value} = 4500 \times (1.032)^4$$

$$(1.032)^4 = 1.13428... \text{ (keep full value)}$$

$$\text{Value} = 4500 \times 1.13428... = 5104.24...$$

Total interest = final value – original:

$$\text{Interest} = 5104.24 - 4500 = 604.24$$

Value = £5104.24; interest earned = £604.24

WORKED EXAMPLE 2 — REVERSE PERCENTAGE

After a 15% increase, a season ticket costs £218.50. Calculate the original price before the increase.

The £218.50 represents 115% of the original, i.e. the original $\times 1.15$. To reverse, divide by the multiplier — do NOT subtract 15%:

$$\text{Original} = \frac{218.50}{1.15}$$

$$\text{Original} = 190.00$$

Common error: taking 15% off £218.50 gives £185.73, which is wrong. Always divide by the multiplier.

Original price = £190.00

WORKED EXAMPLE 3 — REVERSE COMPOUND DEPRECIATION (GRADE 8/9)

A car depreciates by 12% each year. After 2 years it is worth £9 680. Calculate its original value.

Each year the value is multiplied by $(1 - 0.12) = 0.88$. Over 2 years the multiplier is 0.88^2 . The £9 680 is the original $\times 0.88^2$, so reverse by dividing:

$$\text{Original} = \frac{9680}{0.88^2} = \frac{9680}{0.7744}$$

$$\text{Original} = 12\,500$$

Reversing compound change: divide by the multiplier raised to the power of the number of years.

Original value = £12 500

3 Indices & Surds

Even on calculator papers, surd and index questions often demand an exact answer or full algebraic simplification, where a decimal would lose marks. Manipulate symbolically.

KEY FORMULAE

Index laws: $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{mn}$

Negative / fractional: $a^{-n} = \frac{1}{a^n}$ $a^{(1/n)} = \sqrt[n]{a}$ $a^{(m/n)} = (\sqrt[n]{a})^m$

Surds: $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\left(\frac{a}{b}\right)}$

Rationalising: multiply by $\frac{\sqrt{b}}{\sqrt{b}}$ or by the conjugate $(a - \sqrt{b})$

WORKED EXAMPLE 1 — FRACTIONAL AND NEGATIVE INDICES

Evaluate $16^{-3/4}$ as an exact fraction.

A negative index means reciprocal; the fraction $3/4$ means 'fourth root, then cube':

$$16^{-3/4} = \frac{1}{16^{3/4}}$$

$$16^{3/4} = (\sqrt[4]{16})^3 = (2)^3 = 8$$

$$\text{So } 16^{-3/4} = \frac{1}{8}$$

$$16^{-3/4} = 1/8$$

WORKED EXAMPLE 2 — RATIONALISING WITH A CONJUGATE

Rationalise the denominator of $\frac{6}{4 - \sqrt{2}}$ and simplify fully.

Multiply numerator and denominator by the conjugate $(4 + \sqrt{2})$:

$$\frac{6}{4 - \sqrt{2}} \times \frac{4 + \sqrt{2}}{4 + \sqrt{2}} = \frac{6(4 + \sqrt{2})}{(4 - \sqrt{2})(4 + \sqrt{2})}$$

Expand the denominator using the difference of two squares:

$$(4 - \sqrt{2})(4 + \sqrt{2}) = 16 - 2 = 14$$

$$= \frac{6(4 + \sqrt{2})}{14} = \frac{24 + 6\sqrt{2}}{14} = \frac{12 + 3\sqrt{2}}{7}$$

Divide numerator and denominator by the common factor 2.

$$(12 + 3\sqrt{2})/7$$

WORKED EXAMPLE 3 — EXPANDING AND SIMPLIFYING SURDS (GRADE 8/9)

Expand and simplify $(3 + 2\sqrt{5})^2$, giving your answer in the form $a + b\sqrt{5}$.

Treat as $(3 + 2\sqrt{5})(3 + 2\sqrt{5})$ and expand fully:

$$= 3^2 + 2(3)(2\sqrt{5}) + (2\sqrt{5})^2$$

$$= 9 + 12\sqrt{5} + 4 \times 5$$

$$= 9 + 12\sqrt{5} + 20$$

$$= 29 + 12\sqrt{5}$$

Remember $(2\sqrt{5})^2 = 2^2 \times (\sqrt{5})^2 = 4 \times 5 = 20$, not $2\sqrt{5} \times 2 = 4\sqrt{5}$.

$$29 + 12\sqrt{5}$$

4 Quadratic Equations & the Quadratic Formula

Quadratics that do not factorise require the formula or completing the square. The discriminant tells you how many real solutions exist before you solve.

KEY FORMULAE

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$

Discriminant $b^2 - 4ac$: $> 0 \rightarrow$ two roots $\cdot = 0 \rightarrow$ one repeated root $\cdot < 0 \rightarrow$ no real roots

Completing the square: $x^2 + bx + c = (x + \frac{b}{2})^2 - (\frac{b}{2})^2 + c$

WORKED EXAMPLE 1 — QUADRATIC FORMULA

Solve $3x^2 - 7x - 5 = 0$, giving solutions to 3 significant figures.

Identify $a = 3$, $b = -7$, $c = -5$. Substitute into the formula:

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(-5)}}{2(3)}$$

$$x = \frac{7 \pm \sqrt{49 + 60}}{6} = \frac{7 \pm \sqrt{109}}{6}$$

$$\sqrt{109} = 10.4403\dots$$

$$x = \frac{7 + 10.4403}{6} = 2.9067\dots \text{ or } x = \frac{7 - 10.4403}{6} = -0.5734\dots$$

$$x = 2.91 \text{ or } x = -0.573 \text{ (3 s.f.)}$$

WORKED EXAMPLE 2 — COMPLETING THE SQUARE & TURNING POINT

Write $x^2 - 6x + 11$ in the form $(x - p)^2 + q$ and hence state the coordinates of the minimum point.

Halve the coefficient of x ($-6 \rightarrow -3$) and write the squared bracket, then correct the constant:

$$x^2 - 6x + 11 = (x - 3)^2 - (3)^2 + 11$$

$$= (x - 3)^2 - 9 + 11 = (x - 3)^2 + 2$$

The minimum of $(x - 3)^2 + 2$ occurs when the bracket is zero, i.e. $x = 3$, giving $y = 2$:

$$(x - 3)^2 + 2; \text{ minimum point at } (3, 2)$$

WORKED EXAMPLE 3 — FORMING AND SOLVING FROM CONTEXT (GRADE 8/9)

A rectangle has length $(x + 3)$ cm and width $(x - 1)$ cm. Its area is 20 cm^2 . Form an equation and solve it to find x , giving your answer to 3 s.f. Explain why only one solution is valid.

Area = length $\times$ width = 20:

$$(x + 3)(x - 1) = 20$$

$$x^2 + 2x - 3 = 20$$

$$x^2 + 2x - 23 = 0$$

This does not factorise, so use the quadratic formula with $a = 1$, $b = 2$, $c = -23$:

$$x = \frac{-2 \pm \sqrt{4 + 92}}{2} = \frac{-2 \pm \sqrt{96}}{2}$$

$$\sqrt{96} = 9.7980\dots \rightarrow x = 3.899\dots \text{ or } x = -5.899\dots$$

A width of $(x - 1)$ must be positive, so $x > 1$. The negative root gives negative dimensions and is rejected.

$$x = 3.90 \text{ cm (3 s.f.)}; \text{ the negative root is rejected as lengths must be positive}$$

5 Simultaneous Equations (Linear & Quadratic)

Higher papers frequently pair one linear and one quadratic equation, solved by substitution. Linear pairs are solved by elimination.

METHOD SUMMARY

Two linear: eliminate one variable by matching coefficients, then add or subtract.

One linear + one quadratic: rearrange the linear equation, substitute into the quadratic, solve the resulting quadratic, then back-substitute.

WORKED EXAMPLE 1 — ELIMINATION

Solve $4x + 3y = 18$ and $5x - 2y = 11$.

Make the y-coefficients match. Multiply the first by 2 and the second by 3:

$$8x + 6y = 36$$

$$15x - 6y = 33$$

Add the equations to eliminate y:

$$23x = 69 \rightarrow x = 3$$

Substitute $x = 3$ into $4x + 3y = 18$:

$$12 + 3y = 18 \rightarrow 3y = 6 \rightarrow y = 2$$

$$x = 3, y = 2$$

WORKED EXAMPLE 2 — LINEAR AND QUADRATIC

Solve $y = 2x - 1$ and $x^2 + y^2 = 25$.

Substitute the linear equation into the quadratic:

$$x^2 + (2x - 1)^2 = 25$$

$$x^2 + 4x^2 - 4x + 1 = 25$$

$$5x^2 - 4x - 24 = 0$$

Factorise (or use the formula): $5x^2 - 4x - 24 = (5x + \dots) \dots$ — use the formula to be safe:

$$x = \frac{4 \pm \sqrt{(16 + 480)}}{10} = \frac{4 \pm \sqrt{496}}{10}$$

$$\sqrt{496} = 22.2710\dots \rightarrow x = 2.627\dots \text{ or } x = -1.827\dots$$

Back-substitute into $y = 2x - 1$:

$$x = 2.627 \rightarrow y = 4.254; x = -1.827 \rightarrow y = -4.654$$

$$x = 2.63, y = 4.25 \text{ or } x = -1.83, y = -4.65 \text{ (3 s.f.)}$$

6 Vectors

Vector questions typically end with a proof — either that two vectors are parallel, or that three points are collinear (lie on a straight line). Express the required vector as a route through known vectors.

KEY IDEAS

Route method: travel along known vectors; e.g. $AB = AO + OB = -a + b = b - a$

Parallel vectors: one is a scalar multiple of the other, i.e. $p = k \cdot q$ for some number k

Collinear points: if $AB = k \cdot BC$ (sharing point B), then A, B, C lie on a straight line

WORKED EXAMPLE 1 — FINDING A VECTOR VIA A ROUTE

OACB is a parallelogram with $OA = a$ and $OB = b$. M is the midpoint of AC. Express OM in terms of a and b.

In parallelogram OACB, side BC equals OA, so AC is parallel and equal to OB, i.e. $AC = b$. M is the midpoint of AC, so $AM = \frac{1}{2}b$.

Travel from O to A to M:

$$OM = OA + AM = a + \frac{1}{2}b$$

$$OM = a + \frac{1}{2}b$$

WORKED EXAMPLE 2 — COLLINEARITY PROOF

Points have position vectors $OP = 2a + b$, $OQ = 5a + 4b$, $OR = 8a + 7b$. Show that P, Q and R are collinear.

Find PQ and QR using the route method (subtract position vectors):

$$PQ = OQ - OP = (5a + 4b) - (2a + b) = 3a + 3b$$

$$QR = OR - OQ = (8a + 7b) - (5a + 4b) = 3a + 3b$$

Since $PQ = QR$, the vectors are parallel and share the common point Q. Therefore P, Q and R lie on the same straight line.

State the conclusion explicitly — the proof mark requires the worded reason.

$PQ = QR = 3a + 3b \Rightarrow$ parallel & share Q $\Rightarrow$ collinear

WORKED EXAMPLE 3 — VECTOR PROOF (GRADE 8/9)

OABC is a parallelogram with $OA = a$ and $OC = c$. M is the point of intersection of the diagonals OB and AC. Prove using vectors that M is the midpoint of both diagonals.

In parallelogram OABC, AB is equal and parallel to OC, so $OB = OA + AB = a + c$. The midpoint of OB is therefore:

$$OM \text{ (midpoint of OB)} = \frac{1}{2}(a + c)$$

Now consider the midpoint of the other diagonal AC. A general point on AC is $OA + t \cdot AC$, where $AC = OC - OA = c - a$:

$$\text{Point on AC} = a + t(c - a) = (1 - t)a + tc$$

Set this equal to $\frac{1}{2}(a + c) = \frac{1}{2}a + \frac{1}{2}c$ and match the coefficients of a and c:

$$(1 - t) = \frac{1}{2} \rightarrow t = \frac{1}{2} \text{ and } t = \frac{1}{2} \checkmark \text{ (consistent)}$$

Both coefficients give $t = \frac{1}{2}$, so the midpoint of OB is exactly the midpoint of AC. Therefore the diagonals bisect each other at M.

Matching coefficients of independent vectors a and c is the standard technique for vector intersection proofs.

$M = \frac{1}{2}(a + c)$; the diagonals bisect each other ($AM : MC = OM : MB = 1 : 1$)

7 Circle Theorems

Every angle you state must be justified by naming the theorem used — the reasoning carries the marks. Look for radii, tangents, diameters and cyclic quadrilaterals.

THE THEOREMS

Angle at the centre = $2 \times$ angle at the circumference (same arc)

Angle in a semicircle = 90°

Angles in the same segment are equal

Opposite angles of a cyclic quadrilateral sum to 180°

A tangent meets a radius at 90° ; two tangents from a point are equal

Alternate segment theorem: angle between tangent and chord = angle in the alternate segment

WORKED EXAMPLE 1 — CENTRE AND CYCLIC QUADRILATERAL

A, B, C, D lie on a circle, centre O. Angle BOD (at the centre, reflex not used) = 130° . Find angle BAD and angle BCD, giving reasons.

Angle BAD is at the circumference subtending the same arc BD as the centre angle BOD:

$$\text{Angle BAD} = \frac{1}{2} \times 130^\circ = 65^\circ \text{ (angle at centre} = 2 \times \text{angle at circumference)}$$

ABCD is a cyclic quadrilateral, so opposite angles sum to 180° :

$$\text{Angle BCD} = 180^\circ - 65^\circ = 115^\circ \text{ (opposite angles of a cyclic quadrilateral)}$$

Angle BAD = 65° ; angle BCD = 115°

WORKED EXAMPLE 2 — ALTERNATE SEGMENT THEOREM

TA is a tangent to a circle at A. Chord AB makes an angle of 58° with the tangent. Angle ACB is in the alternate segment. State angle ACB, with reason, and find angle ABC if angle BAC = 74° .

By the alternate segment theorem, the angle between tangent and chord equals the angle in the alternate segment:

$$\text{Angle ACB} = 58^\circ \text{ (alternate segment theorem)}$$

Angles in triangle ABC sum to 180° :

$$\text{Angle ABC} = 180^\circ - 74^\circ - 58^\circ = 48^\circ$$

$$\text{Angle ACB} = 58^\circ; \text{ angle ABC} = 48^\circ$$

WORKED EXAMPLE 3 — MULTI-STEP WITH REASONS (GRADE 8/9)

A, B and C are points on a circle, centre O. BD is a tangent to the circle at B. Angle OBC = 32° and OB = OC (radii). Find angle DBC and angle BAC, giving a reason for each step.

Triangle OBC is isosceles (OB = OC are radii), so the base angles are equal:

$$\text{Angle OCB} = \text{angle OBC} = 32^\circ \text{ (base angles of isosceles triangle)}$$

$$\text{Angle BOC} = 180^\circ - 32^\circ - 32^\circ = 116^\circ \text{ (angles in a triangle).}$$

The tangent BD meets radius OB at 90° :

$$\text{Angle DBC} = 90^\circ - 32^\circ = 58^\circ \text{ (tangent } \perp \text{ radius)}$$

Angle BAC is at the circumference subtending arc BC; angle BOC is at the centre on the same arc:

$$\text{Angle BAC} = \frac{1}{2} \times 116^\circ = 58^\circ \text{ (angle at centre} = 2 \times \text{angle at circumference)}$$

$$\text{Angle DBC} = 58^\circ; \text{ angle BAC} = 58^\circ$$

8 Similar Shapes — Length, Area & Volume Scale Factors

A frequent trap: students apply the length scale factor to areas or volumes. Areas scale by the square of the length factor, volumes by the cube.

SCALE FACTOR RULES

Length scale factor = k

Area scale factor = k^2

Volume scale factor = k^3

WORKED EXAMPLE 1 — VOLUME FROM A LENGTH RATIO

Two similar cones have heights 6 cm and 9 cm. The smaller has volume 80 cm^3 . Find the volume of the larger, to 3 s.f.

Length scale factor from small to large:

$$k = \frac{9}{6} = 1.5$$

Volume scale factor = k^3 :

$$k^3 = 1.5^3 = 3.375$$

$$\text{Larger volume} = 80 \times 3.375 = 270 \text{ cm}^3$$

$$270 \text{ cm}^3$$

WORKED EXAMPLE 2 — LENGTH FROM AN AREA RATIO

Two similar shapes have areas 50 cm^2 and 162 cm^2 . The smaller shape has a side of 10 cm. Find the corresponding side of the larger shape.

Area scale factor:

$$k^2 = \frac{162}{50} = 3.24$$

Length scale factor is the square root:

$$k = \sqrt{3.24} = 1.8$$

$$\text{Corresponding side} = 10 \times 1.8 = 18 \text{ cm}$$

18 cm

WORKED EXAMPLE 3 — SURFACE AREA TO VOLUME (GRADE 8/9)

Two similar solids have surface areas in the ratio 9 : 25. The smaller solid has a volume of 54 cm^3 . Find the volume of the larger solid.

Surface area is an area, so the area scale factor is $k^2 = 25/9$ (large : small). Find the length scale factor k :

$$k = \sqrt{\left(\frac{25}{9}\right)} = \frac{5}{3}$$

Volume scale factor is k^3 :

$$k^3 = \left(\frac{5}{3}\right)^3 = \frac{125}{27}$$

$$\text{Larger volume} = 54 \times \frac{125}{27} = 2 \times 125 = 250 \text{ cm}^3$$

Chain the conversions: area ratio $\rightarrow$ take square root for length $\rightarrow$ cube for volume.

250 cm³

9 Bounds & Accuracy

When quantities are rounded, calculations carry a range of possible values. Choosing the correct bound for each operation is the examined skill.

COMBINING BOUNDS

Addition / Multiplication: max uses $UB+UB$ or $UB \times UB$; min uses $LB+LB$ or $LB \times LB$

Subtraction: max = $UB - LB$; min = $LB - UB$

Division: max = $\frac{UB}{LB}$; min = $\frac{LB}{UB}$

(UB = upper bound, LB = lower bound)

WORKED EXAMPLE 1 — UPPER BOUND OF A SPEED

A car travels 240 m (to the nearest 10 m) in 15.6 s (to the nearest 0.1 s). Calculate the upper bound for its average speed, to 3 s.f.

Bounds for distance (nearest 10 m $\rightarrow \pm 5$): $235 \leq d < 245$. Bounds for time (nearest 0.1 $\rightarrow \pm 0.05$): $15.55 \leq t < 15.65$.

Speed = distance $\div$ time. Maximum speed uses the largest distance and the smallest time:

$$\text{Upper bound speed} = \frac{245}{15.55} = 15.756... \text{ m/s}$$

Upper bound = 15.8 m/s (3 s.f.)

WORKED EXAMPLE 2 — DECIDING ACCURACY

Using the data above, give the average speed to a suitable degree of accuracy, justifying your answer.

Lower bound speed uses the smallest distance and largest time:

$$\text{Lower bound} = \frac{235}{15.65} = 15.016... \text{ m/s}$$

Upper bound = 15.756... (from example 1). Both bounds round to 16 m/s to 2 s.f., but differ at 3 s.f. (15.0 vs 15.8).

Since the bounds agree to 2 s.f., the speed can sensibly be stated as 16 m/s (2 s.f.).

16 m/s (2 s.f.) — the value to which both bounds agree

WORKED EXAMPLE 3 — MINIMUM DENSITY (GRADE 8/9)

An object has mass 250 g (to the nearest 5 g) and volume 100 cm³ (to the nearest 5 cm³). Calculate the lower bound for its density, to 3 s.f.

Mass bounds (nearest 5 → ±2.5): $247.5 \leq m < 252.5$. Volume bounds (nearest 5 → ±2.5): $97.5 \leq V < 102.5$.

Density = mass ÷ volume. The minimum density uses the smallest mass and the largest volume:

$$\text{Lower bound density} = \frac{247.5}{102.5} = 2.4146... \text{ g/cm}^3$$

Lower bound density = 2.41 g/cm³ (3 s.f.)

10 Cumulative Frequency, Box Plots & Histograms

Data-handling questions on the calculator papers reward careful reading from graphs and correct use of frequency density for unequal class widths.

KEY RELATIONSHIPS

Median = $\frac{1}{2}n$ th value · **LQ** = $\frac{1}{4}n$ th · **UQ** = $\frac{3}{4}n$ th (read from cumulative frequency)

Interquartile range = UQ – LQ

Frequency density = $\frac{\text{frequency}}{\text{class width}}$ · **Frequency** = freq. density × class width (= bar area)

WORKED EXAMPLE 1 — QUANTILES FROM CUMULATIVE FREQUENCY

A cumulative frequency graph for 80 students rises smoothly. Readings give: at the $\frac{1}{4}$ position the value is 42, at the $\frac{1}{2}$ position 55, at the $\frac{3}{4}$ position 64. Find the median and the interquartile range.

With $n = 80$: median is the 40th value, LQ the 20th, UQ the 60th — read directly from the stated positions:

$$\text{Median} = 55$$

$$\text{LQ} = 42, \text{UQ} = 64$$

$$\text{IQR} = \text{UQ} - \text{LQ} = 64 - 42 = 22$$

Median = 55; IQR = 22

WORKED EXAMPLE 2 — HISTOGRAM FREQUENCY DENSITY

A histogram has a bar covering the class $20 \leq x < 50$ with frequency density 1.4. Find the frequency for this class.

Class width = $50 - 20 = 30$. Frequency = frequency density × class width:

$$\text{Frequency} = 1.4 \times 30 = 42$$

Frequency = 42

WORKED EXAMPLE 3 — ESTIMATED MEAN FROM GROUPED DATA (GRADE 7/8)

The table shows times (minutes) for 32 runners. 0–10: 4 runners; 10–20: 10; 20–30: 12; 30–40: 6. Estimate the mean time.

Use the midpoint of each class as the representative value (5, 15, 25, 35), then compute $\Sigma(fx) \div \Sigma f$:

$$\Sigma(fx) = (5)(4) + (15)(10) + (25)(12) + (35)(6)$$

$$= 20 + 150 + 300 + 210 = 680$$

$$\Sigma f = 4 + 10 + 12 + 6 = 32$$

$$\text{Estimated mean} = \frac{680}{32} = 21.25 \text{ minutes}$$

It is an estimate because the exact values within each class are unknown — the midpoint is assumed. The calculator's STAT mode computes this quickly.

Estimated mean $\approx$ 21.25 minutes

11 Direct & Inverse Proportion

Proportion questions require you to form an equation with a constant k , find k from given values, then use the completed equation. The wording tells you the relationship.

SETTING UP PROPORTION

Directly proportional: $y = kx$ (or $y = kx^2$, $y = kx^3$, $y = k\sqrt{x}$)

Inversely proportional: $y = \frac{k}{x}$ (or $\frac{k}{x^2}$, etc.)

Method: write the equation $\rightarrow$ substitute the known pair to find $k \rightarrow$ rewrite with $k \rightarrow$ answer the question

WORKED EXAMPLE 1 — INVERSE SQUARE PROPORTION (GRADE 7/8)

y is inversely proportional to x^2 . When $x = 3$, $y = 8$. Find y when $x = 6$.

Inverse proportion to x^2 means:

$$y = \frac{k}{x^2}$$

Substitute $x = 3$, $y = 8$ to find k :

$$8 = \frac{k}{9} \rightarrow k = 72$$

Rewrite the equation and substitute $x = 6$:

$$y = \frac{72}{x^2} \rightarrow y = \frac{72}{36} = 2$$

$y = 2$

WORKED EXAMPLE 2 — DIRECT PROPORTION TO A CUBE (GRADE 8)

The mass M of a sphere is directly proportional to the cube of its radius r . A sphere of radius 2 cm has mass 60 g. Find the mass of a sphere of radius 5 cm.

Direct proportion to r^3 :

$$M = k \cdot r^3$$

Substitute $r = 2$, $M = 60$:

$$60 = k \times 8 \rightarrow k = 7.5$$

Find M when $r = 5$:

$$M = 7.5 \times 5^3 = 7.5 \times 125 = 937.5 \text{ g}$$

$M = 937.5 \text{ g}$

WORKED EXAMPLE 3 — INTERPRETING THE EFFECT OF A CHANGE (GRADE 9)

y is inversely proportional to the square of x. If x is doubled, what happens to y?

Start with $y = k/x^2$. Replace x with $2x$ to model doubling:

$$y_{\text{new}} = \frac{k}{(2x)^2} = \frac{k}{4x^2} = \frac{1}{4} \times \frac{k}{x^2}$$

Since k/x^2 was the original y, the new value is one quarter of the original.

y becomes one quarter ($\frac{1}{4}$) of its original value

12 Iteration

Iteration finds approximate solutions to equations that cannot be solved exactly. Rearrange to $x = g(x)$, then repeatedly feed each output back in using the ANS key.

METHOD

Use the given iterative formula $x_{n+1} = g(x_n)$

Start from the given x_0 ; compute $x_1, x_2, x_3 \dots$ on your calculator

Efficient method: type the formula using ANS, then press = repeatedly

WORKED EXAMPLE 1 — APPLYING AN ITERATIVE FORMULA (GRADE 8)

Using $x_{n+1} = \sqrt[3]{5x_n + 2}$ with $x_0 = 2$, find x_1, x_2 and x_3 , each to 5 decimal places.

Substitute repeatedly. Enter 2, then on the calculator type the cube root of $(5 \cdot \text{ANS} + 2)$ and press = each time:

$$x_1 = \sqrt[3]{5 \times 2 + 2} = \sqrt[3]{12} = 2.28943$$

$$x_2 = \sqrt[3]{5 \times 2.28943 + 2} = \sqrt[3]{13.44715} = 2.37799$$

$$x_3 = \sqrt[3]{5 \times 2.37799 + 2} = \sqrt[3]{13.88995} = 2.40381$$

$x_1 = 2.28943, x_2 = 2.37799, x_3 = 2.40381$

WORKED EXAMPLE 2 — SHOWING THE ROOT AND ACCURACY (GRADE 9)

The iteration above converges to a root of $x^3 - 5x - 2 = 0$. Continuing gives $x_4 = 2.41124$. State the root to 2 decimal places and explain how the iterates justify it.

Successive iterates are 2.37799, 2.40381, 2.41124 — they are getting closer together and stabilising around 2.41.

Since x_3 and x_4 both round to 2.41 to 2 d.p., the sequence has converged to this accuracy:

Root ≈ 2.41 (2 d.p.)

To prove a root lies in an interval, you can also show the function changes sign across it (e.g. $f(2.40) < 0$ and $f(2.42) > 0$).

Root ≈ 2.41 (2 d.p.)

13 Probability — Tree Diagrams & Conditional

The hardest probability questions involve selection 'without replacement', where the second probability depends on the first. Multiply along branches, add between outcomes.

KEY RULES

AND (along branches): multiply the probabilities

OR (between separate outcomes): add the results

Without replacement: the denominator (and matching numerator) decreases by 1 for the second pick

WORKED EXAMPLE 1 — WITHOUT REPLACEMENT (GRADE 7/8)

A bag contains 5 red and 3 blue counters. Two are taken at random without replacement. Find the probability that both are red.

First pick red: 5 out of 8. After removing one red, 4 red remain out of 7. Multiply along the 'red then red' branches:

$$P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$$

$$P(\text{both red}) = 5/14$$

WORKED EXAMPLE 2 — 'ONE OF EACH' (GRADE 8)

Using the same bag (5 red, 3 blue, two taken without replacement), find the probability of one red and one blue, in any order.

Two routes give one of each: red-then-blue, or blue-then-red. Compute each and add:

$$P(R \text{ then } B) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$$

$$P(B \text{ then } R) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$$

$$\text{Total} = \frac{15}{56} + \frac{15}{56} = \frac{30}{56} = \frac{15}{28}$$

$$P(\text{one of each}) = 15/28$$

WORKED EXAMPLE 3 — CONDITIONAL PROBABILITY (GRADE 9)

Using the same bag, given that the first counter taken (and kept) was red, find the probability that the second is also red.

This is a conditional probability — we are told the first was red, so the bag now holds 4 red and 3 blue (7 counters):

$$P(\text{second red} \mid \text{first red}) = \frac{4}{7}$$

Conditioning on the first event simply means working from the reduced bag; no extra formula is needed here.

$$P(\text{second red} \mid \text{first red}) = 4/7$$

14 Arc Length, Sector Area & Segments

Sector calculations are pure fractions of a full circle. The area of a segment combines a sector with a triangle, often needing the $\frac{1}{2}ab \cdot \sin C$ area formula.

KEY FORMULAE

$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r \quad \text{Sector area} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Area of segment} = \text{sector area} - \text{area of triangle} = \frac{\theta}{360} \pi r^2 - \frac{1}{2} r^2 \cdot \sin \theta$$

WORKED EXAMPLE 1 — ARC AND SECTOR (GRADE 7)

A sector of a circle has radius 8 cm and angle 135° . Find its arc length and area, to 3 s.f.

Arc length is the fraction $\theta/360$ of the full circumference:

$$\text{Arc} = \frac{135}{360} \times 2 \times \pi \times 8 = 0.375 \times 50.265... = 18.849... \text{ cm}$$

Sector area is the same fraction of the full circle area:

$$\text{Area} = \frac{135}{360} \times \pi \times 8^2 = 0.375 \times 201.06... = 75.398... \text{ cm}^2$$

Arc length = 18.8 cm; sector area = 75.4 cm² (3 s.f.)

WORKED EXAMPLE 2 — AREA OF A SEGMENT (GRADE 9)

For the same sector (radius 8 cm, angle 135°), find the area of the segment between the chord and the arc, to 3 s.f.

Segment = sector area – area of the triangle formed by the two radii and the chord. The triangle has two sides of 8 cm with included angle 135° :

$$\text{Triangle area} = \frac{1}{2} \times 8 \times 8 \times \sin 135^\circ = 32 \times 0.70710... = 22.627... \text{ cm}^2$$

Subtract from the sector area found previously:

$$\text{Segment} = 75.398... - 22.627... = 52.771... \text{ cm}^2$$

Segment area = 52.8 cm² (3 s.f.)

15 Algebraic Fractions

Simplifying and combining algebraic fractions relies on factorising first. Look for common factors and the difference of two squares before cancelling.

APPROACH

Simplify: factorise numerator and denominator fully, then cancel common factors

Add / subtract: use a common denominator, exactly as with numerical fractions

Solve: multiply through by the denominator to clear fractions, then solve

WORKED EXAMPLE 1 — SIMPLIFYING (GRADE 8)

Simplify $\frac{x^2 - 9}{x^2 + x - 6}$ fully.

Factorise the numerator (difference of two squares) and the denominator (quadratic):

$$\text{Numerator: } x^2 - 9 = (x + 3)(x - 3)$$

$$\text{Denominator: } x^2 + x - 6 = (x + 3)(x - 2)$$

Cancel the common factor $(x + 3)$:

$$\frac{(x + 3)(x - 3)}{(x + 3)(x - 2)} = \frac{x - 3}{x - 2}$$

$(x - 3)/(x - 2)$

WORKED EXAMPLE 2 — ADDING ALGEBRAIC FRACTIONS (GRADE 9)

Express as a single fraction: $\frac{3}{x+1} + \frac{2}{x-2}$.

The common denominator is $(x+1)(x-2)$. Rewrite each fraction over it:

$$= \frac{3(x-2)}{(x+1)(x-2)} + \frac{2(x+1)}{(x+1)(x-2)}$$

Combine the numerators and expand:

$$\begin{aligned} &= \frac{3(x-2) + 2(x+1)}{(x+1)(x-2)} = \frac{3x-6+2x+2}{(x+1)(x-2)} \\ &= \frac{5x-4}{(x+1)(x-2)} \end{aligned}$$

Leave the denominator in factorised form unless asked to expand it.

$$(5x-4) / [(x+1)(x-2)]$$